

Subjectivity-Intersection Braid Quantum Gravity

A Constructive Refutation of Objectivist Primacy with a Source-Derived 1 + 3 Carrier and a Two-Principle Conditional Einstein Equation

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Claim status. This paper presents Subjectivity-Intersection Braid Quantum Gravity as a candidate theory, not as a completed theory of quantum gravity. A target-free observer quotient produces a minimal $1 \oplus 3$ carrier without supplying the numeral four or an ambient four-manifold; one explicit Cauchy-completion principle turns that carrier into a real affine continuum. Exact finite theorems also generate metric-affine dynamics, source-native actions, local quantum evolution, and Noether identities. The historical four-law continuum packet reduces by dependency to two explicit principles: causal geometric realization (CGR) and marked matter realization (MMR). Under CGR and MMR, a continuum metric-affine action yields $G_{\mu\nu} = (3/5)T_{\mu\nu}$ as an Euler–Lagrange equation. CGR and MMR are not yet derived from the Braid source; continuum-field quantization and empirical validation remain open.

Abstract

We formulate *Subjectivity-Intersection Braid Quantum Gravity* as a candidate theory in which quantum kinematics, spacetime geometry, and gravitational response arise from one standpoint-bearing pointed-braid source rather than being postulated as independent primitives. Here subjectivity denotes a mathematically or physically located standpoint of access and transport, not consciousness. Distinct source standpoints remain nonidentical, while their Braid-mediated intersection generates a covariant structure shared across changes of presentation.

The source is quantum in the algebraic sense: a noncommutative matrix-algebra tower carries states, unitary pointed-braid transport, and Yang–Baxter-coherent evolution. Its observer quotient is $Q_{\text{obs}} = V_4$. Requiring a real readout to contain the common order unit, separate the four observer states, and respect the full intrinsic affine action forces the complete function carrier. It splits canonically as one constant direction plus a three-dimensional mean-zero representation. Thus dimension four and the 1 + 3 type are outputs of a stated observer-completeness principle rather than requested target dimensions. Under Observer Uniform Cauchy Closure, the rational carrier completes without new directions to $W = \mathbb{R}^{V_4}$, whose affine space $\text{Aff}(W)$ is a smooth four-dimensional carrier.

Exact finite certificates further generate a nonzero covariant symmetric tensor carrier, source-native coframes and holonomies, finite metric-affine gravity equations with torsion and nonmetricity, graph-local propagation, a source-local GKSL evolution, nonzero connection curvature, first-order curvature and defect actions, and discrete Noether identities. These constructions regenerate through all declared finite refinement stages without stage-dependent fitting. The one-dimensional source path is not labelled as three-dimensional physical space: it indexes operator locality, while the source-derived 1 + 3 representation supplies the local event type.

We then separate pointwise arrival from variational derivation. The earlier four bridge laws are not independent. Their present content reduces to two explicit realization principles. CGR maps the completed observer carrier, source-local net, temporal affine Cartan link, and finite Palatini cells to local metric-affine fields with a refinement-compatible continuum first variation. MMR maps the source G_1 Gibbs/KMS seed and doubled matter response to one local metric-only matter action, and routes the source rank-three marked sector through the common trace. Under CGR and MMR the infrared action

$$S_{\text{IR}} = \frac{1}{2\kappa_B} \int d^4x \sqrt{-g} g^{\mu\nu} R_{(\mu\nu)}(\Gamma) + S_m[g, \Gamma, \psi]$$

yields, in the long-wavelength, low-curvature, matter-hypermomentum-free sector,

$$G_{\mu\nu}(g) = \kappa_B T_{\mu\nu}, \quad \kappa_B = \frac{3}{5},$$

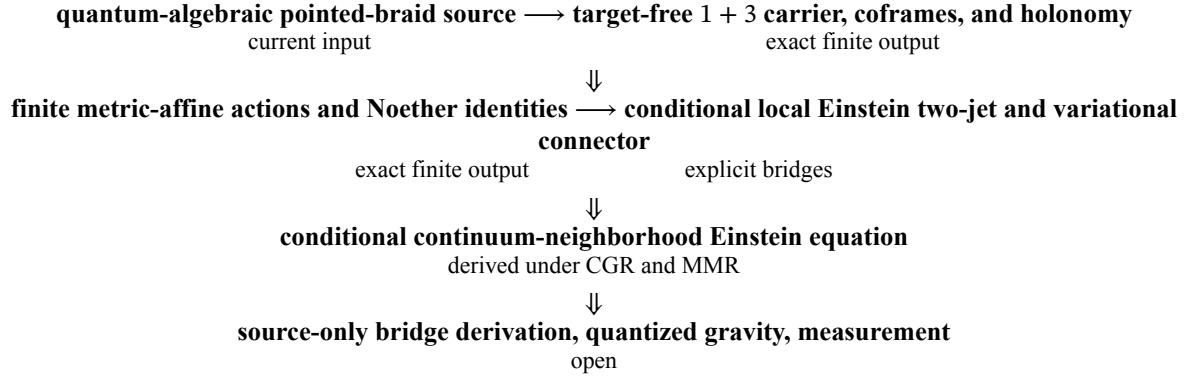
as an Euler–Lagrange field equation on a continuous spacetime neighborhood rather than as a fit at one point.

This result strengthens the constructive refutation of *objectivist primacy*: neither the shared tensor carrier nor its numerical four-dimensional type must be installed as a completed objective background. Objectivity is retained as covariance and inter-presentation stability while its ontological priority is denied as a universal necessity. The construction is nevertheless not a completed quantum-gravity theory. CGR and MMR remain explicit non-Braid-derived principles, the tested temporal four-flow distribution is noninvolutive, and the continuum gravitational field has not been quantized. No ultraviolet completion or empirical gravitational identification is claimed. The result is an explicit and falsifiable Braid-based quantum-gravity candidate grounded in subjectivity intersection.

Keywords: Subjectivity-Intersection Braid Quantum Gravity; quantum gravity candidate; pointed braid; generated objectivity; objectivist primacy; continuum Einstein equation; metric-affine gravity; emergent spacetime; Creation Principle; Yang–Baxter coherence.

1 The Braid-first question

The proposal is intentionally ambitious: begin from one standpoint-bearing pointed-braid source and ask how much of the architecture normally assigned separately to quantum theory, spacetime, and gravity can be recovered as outputs of one generative law. The present evidence establishes the middle finite part of that programme, not its endpoints. Its logical status is summarized by



The research wager is that CGR and MMR can ultimately be supplied by strengthening the same source law rather than attaching an unrelated spacetime or gravitational theory afterward. The present result no longer stops at a pointwise Einstein-compatible geometry: after stating those two principles openly, it reaches a variational Einstein equation on a continuous neighborhood. The exact source results and the conditional continuum theorem are kept separate throughout.

Modern physics succeeds by extracting quantities that do not depend on arbitrary coordinates, gauges, or laboratory presentations. The methodological value of objectivity is not in dispute here. The disputed step is ontological: must the invariant structure be there first, independently of every standpoint, with perspectives entering only as partial descriptions of it? Call that stronger commitment *objectivist primacy*. It is stronger than scientific reproducibility, stronger than observer agreement, and stronger than covariance.

The distinction matters because invariance alone does not specify the order of construction. One may begin with an objective object and derive its appearances. Alternatively, one may begin with distinct standpoints and derive a public invariant by quotienting only their representational redundancy. Both routes can end with an invariant. They disagree about what is primitive.

This paper develops the second route as the proven finite core and conditional continuum extension of Subjectivity-Intersection Braid Quantum Gravity. The mathematical work is motivated by Subjectivity Intersection Mathematics and its pointed-unitary-braid realization [16, 14]. Its first aim is not to replace the predictions of established physics. It is to establish that the same source can generate classical objective structure and finite gravity dynamics, and can support a conditional continuum Einstein law, so that the primacy of objectivity is not forced by exactness, covariance, reproducibility, or stable shared structure.

The proposal is close enough to relational quantum mechanics, quantum reference frames, QBism, perspectival realism, and structural realism that careful separation is essential. Relational quantum mechanics denies an observer-independent quantum state and treats facts as relational [10]. Quantum-reference-frame formalisms construct transformations among descriptions associated with physical systems and generalize covariance without an external absolute frame [4, 11]. QBism treats the quantum state as an agent's personal probabilistic assignment rather than an objective property [3]. Perspectival realism studies reliable scientific knowledge across situated perspectives [7, 8]. Structural realism and invariantism identify objective content with transformation-invariant structure [13, 6]. The present proposal shares relational and perspectival motivations, but makes a different technical claim: an objective tensor carrier is an *output* of the maintained distinction and intersection of two source standpoints, and the same output is regenerated at every finite stage of a nonterminal braid tower.

2 Three claims that must not be conflated

2.1 Subjectivity is not consciousness

In this paper, a *subjectivity* is a located mode of access, transport, or presentation that cannot simply be deleted while preserving the construction. It may be represented by a physical reference system, a source prefix, or a typed algebraic standpoint. Nothing in the mathematics attributes sensation, awareness, phenomenal experience, intention, or agency to a braid, a photon, a clock, or an algebra. In particular, *Relative Subjectivity is not an individual subject*: it is not defined by a person, organism, brain, mind, or conscious observer. It names the non-personal relative mode associated, in the proposed physical interpretation, with time, clocking, ordering, and comparison. Human and other biological subjects may instantiate highly organized forms of relative standpoint, but they do not constitute or originate Relative Subjectivity.

This usage is ontological rather than psychological. It says that a physical description is made from somewhere in the structure, and that the difference among such “somewheres” can be constitutive rather than merely epistemic. On the proposed ontology, Relative Subjectivity therefore belongs to physical reality prior to life: clock-like relative ordering and standpoint difference do not begin with humanity, and would obtain in a prebiotic universe. This temporal priority is an ontological interpretation of the physical identification, not a theorem of the present Braid certificates. A separate bridge axiom and separate evidence would be required to connect this notion to consciousness.

2.2 The formal generated-objectivity thesis

Let A and B denote distinct typed source standpoints and let C denote the structure produced from their intersection after allowable presentation dependence has been removed.

Definition 2.1 (Generated objectivity). A carrier $C(A, B)$ exhibits *generated objectivity* if:

1. it is invariant under internal changes of presentation of A and B ;
2. it obeys the appropriate covariance law under a change of ambient description;
3. exchanging the two standpoints does not change the joint carrier;
4. collapsing either standpoint changes the carrier;

5. one source-native law regenerates the carrier through the declared refinement stages without stage-dependent fitting.

Conditions 1–3 provide public stability. Condition 4 prevents “objectivity” from being obtained by deleting the very distinction that was claimed to generate it. Condition 5 makes the carrier a repeated result of a law rather than a one-time finite artifact.

2.3 The ontological Creation Principle

The author’s pre-spacetime interpretation uses the following names:

$$\begin{aligned} A &= \text{light} / \text{Absolute Subjectivity}, & B &= \text{time-clock} / \text{Relative Subjectivity}, \\ C &= \text{space} / \text{Intersection Subjectivity}. \end{aligned}$$

Here B does not denote a human observer or an individual perspective. It denotes an impersonal relative pole whose proposed physical realization is time or clock relation. Accordingly, this ontology places Relative Subjectivity before persons, organisms, and consciousness; its existence is not contingent on the later appearance of humanity. The labels are not premises of the tensor calculation. They constitute an interpretation map from the formal roles to a proposed ontology.

Hypothesis 2.2 (Creation Principle). Objective physical structure is not an independently completed container in which subjectivities later appear. Distinct subjectivities remain nonidentical. Through their intersection they generate a shareable structure C . Re-entry at later stages regenerates rather than merely transports that structure.

The formal result below supports this hypothesis by providing an exact model of its structural grammar. It does not establish the interpretation map to light, time, and physical space. The map becomes a physical theory only when it supplies dynamics, scale, operational readouts, and empirical discrimination.

2.4 The exact target of the counterexample

To avoid attacking a vague “objectivism,” define the target precisely.

Definition 2.3 (Objectivist Primacy Thesis, OP). Every mathematically admissible construction of a stable, intersubjectively shareable, transformation-respecting physical carrier must take that objective carrier, or an equivalent perspective-neutral carrier, as primitive. Distinct standpoints can describe it but cannot generate it.

OP is a universal necessity claim. It is not identical to scientific realism, mind-independent reality, ordinary experimental objectivity, or the claim that invariant quantities are especially informative. One countermodel meeting the advertised conditions is sufficient to refute OP as stated. It is not sufficient to prove that the countermodel is realized in nature.

3 A source-first pointed-braid setting

3.1 Finite source, nonterminal law

Let $\mathcal{A}_n = M_{5^n}(\mathbb{C})$ denote the finite matrix algebra at stage n , with the refinement

$$j_n : \mathcal{A}_n \longrightarrow \mathcal{A}_{n+1}, \quad j_n(X) = X \otimes I_5. \quad (1)$$

Every stage is finite. The sequence has no terminal stage. This distinction is important: a nonterminal tower is not, by itself, a continuum, an infinite-mode quantum field, or physical spacetime. It supplies an exact setting in which one law can be tested at arbitrarily many finite stages.

The source is pointed and braided: it carries a distinguished state and typed unitary braid transports satisfying the declared Yang–Baxter, insertion, and composition laws. The all-orders endpoint-consistency theorem and the positive bilateral response operators were established previously [14]. The present paper uses that source theorem but does not infer physical subjectivity from it.

The computational realization contains eight signed source sectors. Each sector supplies the primitive source packet

$$Q_2 = (R, H, F, P), \quad (2)$$

where the symbols denote the registered relation and source fields, not independently fitted target geometry. Three selected signed-braid generators act as commuting involutions on the certified orbit. From one four-state source slice they generate the opposite slice. Exact checks recover all 12 declared source updates, signed relations, clock fields, and associated source covectors; the identity-map comparator fails in all 12 cases.

3.2 Source-native geometry before the present carrier

The relevant prior source chain is

$$(R, H, F, P) \longrightarrow \text{signed-braid source orbit} \longrightarrow \text{affine split-frame geometry} \longrightarrow \text{coframe and holonomy}. \quad (3)$$

The end-to-end constructor uses no stored future geometry. It reconstructs the connection data, split-frame defect, current term, affine edge, future spatial edges, and future faces from the predicted source relation. In the certified fixed orbit it recovers 12/12 source-slice updates, 24/24 affine edges, 24/24 future spatial edges, and 6/6 future faces exactly. Removing either of two required source sectors breaks the corresponding controls.

This result removes one important circularity: the geometry used below is not inserted as a target answer and then redescribed as braid-generated. It is the output of the registered finite source constructor. The constructor remains global on the fixed finite orbit rather than a strictly local Cauchy law. Version 0.5 separately removes the numerical ambient-dimension input at the carrier layer, as follows.

3.3 A target-free 1 + 3 carrier and its affine completion

Let the source-derived observer quotient be $Q_{\text{obs}} = V_4$, with intrinsic affine action

$$\text{AGL}(2, 2) \cong S_4.$$

Start over the rational source layer and let $W_{\mathbb{Q}} = \mathbb{Q}^{V_4}$ be the space of observer readouts. Impose three explicit admissibility conditions: the readout contains the constant order unit, separates all four quotient states, and is invariant under the full intrinsic affine action. The S_4 permutation commutant is spanned by I and the all-ones matrix J . Its equivariant idempotents have ranks 0, 1, 3, 4. Unitality excludes ranks 0 and 3, while point separation excludes rank 1. Therefore the minimal admissible readout is the full carrier, without supplying a target dimension:

$$W_{\mathbb{Q}} = \mathbb{Q}\mathbf{1} \oplus W_0, \quad \dim(\mathbb{Q}\mathbf{1}, W_0) = (1, 3). \quad (4)$$

The three-space W_0 is the irreducible mean-zero representation. All 24 affine observer actions preserve this split.

The finite Braid data do not by themselves contain the real numbers as a completed scalar field. We therefore state one completion principle explicitly.

Assumption 3.1 (Observer Uniform Cauchy Closure). Observer readouts and the marked clock direction are closed under uniform Cauchy limits without adding independent observer directions.

Under Assumption 3.1, scalar completion gives

$$W = \mathbb{R}^{V_4} = \mathbb{R}\mathbf{1} \oplus W_{0,\mathbb{R}}, \quad \dim W = 4, \quad (5)$$

with the same 1 + 3 split. Every finite-dimensional real vector space canonically determines an affine smooth carrier $\text{Aff}(W)$: choosing an origin gives a global chart, changing that origin is only a translation, and

$$T_x \text{Aff}(W) \cong W \quad (6)$$

for every x . Thus neither the numeral four nor a separate ambient four-manifold is a primitive mathematical input. A declared cup/null composition additionally gives the dimensionless Lorentz form

$$g_D = \frac{16}{75} \text{diag}(-3, 1, 1, 1), \quad (7)$$

whose four evaluation rays are null. Physical rods, seconds, nonconstant curvature, and the identification of points of $\text{Aff}(W)$ with physical events do not follow from this algebraic completion.

The source path and the carrier have different roles. The path indexes inclusions and disjoint supports in the operator net; it is not renamed as three spatial axes. The three relative directions in Eq. (4) are internal observer-readout directions. Identifying $\text{Aff}(W)$ with a physical causal event carrier and soldering the operator net and finite Cartan fields to its regions is part of CGR below. This makes the remaining physical bridge explicit while removing a hidden four-dimensional background choice.

3.4 The source also closes a finite gravity equation

The source-native construction does not stop at a static geometry. On each certified edge (m, d) it obeys the exact first-order coframe equation

$$E_{m \oplus d} = C_{m,d} E_m R_d + K_{m,d} [C, j, F], \quad (8)$$

with the second covariant face closure

$$D_C K = F_C E. \quad (9)$$

Here $C_{m,d}$ is the source-generated ECDR/BSFG connection, R_d reverses the crossed generator orientation, j is the IBC current, F is the affine face curvature recomputed from (C, j) , and $K[C, j, F]$ is the coframe source that cannot be produced by invertible transport alone.

On the fixed eight-source orbit, Eq. (8) recovers 24/24 future coframes, 24/24 nonzero source terms, and all future ranks exactly; Eq. (9) holds on 12/12 faces and is nonzero on 10/12. Transport without K fails on all 12 rank-changing edges, while zeroing the curvature fails on all 40 nonzero-curvature columns. The source term is therefore not a removable decoration of a stored geometry.

For the induced spatial Gram matrix $G_m = E_m^\top g_m E_m$, the same update yields the finite metric-affine equation

$$\boxed{G_n = R_e^\top G_m R_e + \mathcal{N}_{e,E} + \mathcal{L}_{e,K} + \mathcal{Q}_{e,K}} \quad (10)$$

with

$$\mathcal{N}_{e,E} = R_e^\top E_m^\top (C_e^\top g_n C_e - g_m) E_m R_e, \quad (11)$$

$$\mathcal{L}_{e,K} = (C_e E_m R_e)^\top g_n K_e + K_e^\top g_n (C_e E_m R_e), \quad (12)$$

$$\mathcal{Q}_{e,K} = K_e^\top g_n K_e. \quad (13)$$

All three terms are nonzero and separately necessary on 24/24 edges. Thus the connection's nonmetricity, the source-linear response, and the finite source-quadratic response cannot be collapsed into one another. This is a source-generated finite gravitational dynamics in the declared model. Its packet lifts by one fixed law to every finite braid stage, but only as a stage-compatible cylinder section: it does not yet generate new local spacetime cells.

Equations (8)–(10) are not themselves the Einstein equation $G_{\mu\nu} = \kappa T_{\mu\nu}$. Calling them finite Braid gravity equations records their exact source–geometry dynamics while preserving that distinction. The next construction gives a direct but explicitly conditional local connection to the Einstein equation.

3.5 A conditional source-first local Einstein completion

An initial inverse test showed that the rank-ten UB476 response map can reconstruct a restricted Einstein target to relative error 3.69×10^{-15} while preserving trace and the registered G_1 energy. Because that construction solved backward from the Einstein tensor, it established capacity but not noncircular selection. A stricter comparison then produced an exact no-go: the independently fixed UB476 creation response has nonzero 01, 02, and 03 components, whereas the restricted ultrastatic target sets all three to zero. Even the best unrestricted scalar fit leaves a relative residual of approximately 0.745.

The successful construction therefore reverses the direction. Let $\Theta_{\mu\nu}$ denote the symmetric ten-component UB476 response fixed before any Einstein target is read. Under the three declared bridges

- (i) $\Theta_{\mu\nu}$ is typed as a local gravitational source tensor,
- (ii) the independently fixed marked/full source-rank ratio 3/5 is used as the coupling, without fitting it to an Einstein target, and
- (iii) the source-free Weyl tensor is set to zero as the minimal infrared completion,

the source first fixes

$$G_{\mu\nu}(p) = \frac{3}{5} \Theta_{\mu\nu} \quad (14)$$

at a base point p . Four-dimensional trace reversal determines the Ricci tensor and scalar curvature; zero Weyl curvature then determines an algebraic Riemann tensor. A standard normal-coordinate construction produces a smooth quadratic Lorentz metric with the required two-jet at p [12].

Every step was checked in exact arithmetic. The Riemann tensor has the two pair antisymmetries, pair-exchange symmetry, and algebraic Bianchi identity; its Ricci and Einstein contractions reproduce Eq. (14); reconstruction from the metric two-jet returns the same curvature. At unit formal amplitude it has 136 nonzero components. The nonzero 0i source components are not discarded: they generate 93 nonzero temporal-derivative entries and 84 nonzero mixed-component entries of the metric two-jet. The base metric is nondegenerate with Lorentz signature, so the quadratic metric retains that signature in a sufficiently small neighborhood. The same source packet and construction apply in all eight signed sectors.

This is a genuine source-first Einstein connection: the source is fixed before the geometry, and the resulting metric satisfies the Einstein relation exactly at p . It is also conditional. The numerical ratio 3/5 is source-derived rather than manually tuned, but interpreting that ratio as a gravitational coupling, typing the response as stress, and choosing a zero-Weyl completion are bridge assumptions. Differential Bianchi propagation, a neighborhood solution of the Einstein PDE, Cauchy dynamics, physical units, and empirical gravity are not obtained. Moreover, the independently certified Braid curvature does not yet select the same complete Riemann tensor: the first coverage audit found only one of the six required bivector slots in a common Levi–Civita carrier, and subsequent audits localized the remaining obstruction to placing all temporal and spatial curvature records in one $\text{End}(T)$ family. This is underdetermination, not a computed disagreement with Eq. (14).

3.6 From pointwise arrival to a conditional continuum field equation

The preceding construction is an *arrival* result: it builds a metric two-jet whose Einstein tensor has the required value at one point. A field-equation *derivation* is stronger. It must define an off-shell action on a neighborhood, vary the admissible fields independently, and obtain the equation for the allowed solutions rather than construct one compatible jet.

The v0.4 formulation listed four bridge laws: temporal soldering, continuum/infrared convergence, a matter/Ward law, and common normalization. Subsequent dependency audits show that these are not four independent principles. Temporal soldering and continuum convergence are two components of one source-to-geometry realization. Common normalization is irrelevant in vacuum and, with matter, is part of the matter-action provenance rather than a fourth gravitational law. The minimal current packet is therefore:

Assumption 3.2 (Causal geometric realization, CGR). There is a refinement-compatible realization from the completed carrier $\text{Aff}(W)$, the source-path local net, the temporal $\text{Aff}(V_4)$ Cartan link, and the finite Hodge–Palatini cells to local metric-affine fields and regions such that:

- (i) net inclusions and overlaps map to local regions; causal disjointness and tangent/cotangent typing are preserved;
- (ii) the temporal Cartan link and the finite Palatini cells are natural under refinement; and
- (iii) the localized sums admit the uniform compactness and compactly supported first variations required by the continuum Palatini limit.

Assumption 3.3 (Marked matter realization and routing, MMR). The source G_1 Gibbs/modular response and the doubled density–polarization functional realize one local covariant metric-only matter action on the CGR geometry. The same source trace routes the full sector to gravity and the source rank-three marked sector to matter before target comparison. Consequently,

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta S_m}{\delta g^{\mu\nu}}, \quad \kappa_B = \frac{3}{5}, \quad (15)$$

the on-shell Ward identity holds, and matter hypermomentum vanishes in this selected metric-only effective class.

Under CGR and MMR the working infrared action is

$$S_{\text{IR}}[g, \Gamma, \psi] = \frac{1}{2\kappa_B} \int d^4x \sqrt{-g} g^{\mu\nu} R_{(\mu\nu)}(\Gamma) + S_m[g, \Gamma, \psi]. \quad (16)$$

Metric or coframe variation gives

$$R_{(\mu\nu)}(\Gamma) - \frac{1}{2} g_{\mu\nu} R(\Gamma) = \kappa_B T_{\mu\nu}. \quad (17)$$

The connection equation gives $\Gamma = \Gamma_{\text{LC}}(g)$ in the Lorentz branch and Levi–Civita modulo a projective one-form in the metric-affine branch. The projective shift changes only the antisymmetric Ricci tensor, so it cancels from $R_{(\mu\nu)}$, the Palatini scalar, and Eq. (17). Substitution, followed by the MMR routing, yields

$$\boxed{G_{\mu\nu}(g) = \frac{3}{5} T_{\mu\nu}} \quad (18)$$

as an Euler–Lagrange equation on a continuous spacetime neighborhood.

Theorem 3.4 (Two-principle conditional continuum Einstein theorem). *The exact finite source results of this paper, together with CGR and MMR, imply the continuum-neighborhood field equation (18) in the long-wavelength and low-curvature sector of the selected metric-only matter class. This is a conditional variational derivation and is strictly stronger than the pointwise arrival result (14).*

The theorem does not hide the remaining gap. Neither CGR nor MMR is presently derived in full from the pointed-braid source. The finite side nevertheless reaches farther than v0.4 recorded: a source-derived rank-four metric-response carrier persists through the inductive tower; the scalar response is integrated by the canonical Reynolds-projection GKSL semigroup; the three relative directions are inner-derivation controls; and their sum is a single CPTP hybrid evolution. On the source-path interval net this evolution is strictly local on its marked support and compatible with right-tail refinement. These results supply algebraic locality and evolution, not the CGR localization functor.

The current four response directions are locally independent but not four commuting coordinate flows. Their first Lie-bracket closure exits the rank-four span. This prevents a first-class interpretation of those particular flows, but it does not obstruct the single hybrid GKSL generator or the conditional Palatini variation. MMR also remains nontrivial: the finite G_1 Gibbs state supplies an exact algebraic KMS seed, and the rank ratio 3/5 is structural rather than manually fitted, but a continuum doubled α -metric realization and the marked gravity–matter routing are still explicit requirements.

The two-principle count is minimal relative to the audited source inventory. CGR does not supply a matter action or its relative normalization; MMR does not supply event localization, curvature, or continuum Palatini variation. No common source parent functional currently generates both, and independent nonzero rescalings of the separate gravity and matter actions preserve their component equations while changing their relative coupling. Calling both principles one law would therefore package assumptions rather than derive them.

4 From finite updates to source-generated locality

4.1 A finite source-graph local net

At stage n , let G_n be the graph whose vertices are tensor factors and whose edges join adjacent noncommuting Braid generators. In every certified signed sector, distant generators commute exactly and adjacent triples obey the Yang–Baxter relation. For a vertex set U , define

$$\mathfrak{A}_n(U) = \text{End}((\mathbb{C}^5)^{\otimes U}) \otimes I_{U^c}. \quad (19)$$

Then $U \subseteq V$ implies $\mathfrak{A}_n(U) \subseteq \mathfrak{A}_n(V)$, disjoint supports commute, and a depth- t adjacent-Braid update expands support by at most graph distance t . The refinement j_n is strict and unital. These are exact finite isotony, locality, and finite-propagation statements on a source-generated graph. They are not yet Haag–Kastler locality on a Lorentzian manifold, and graph distance is not identified with physical distance.

The positive Braid generator candidate

$$L_n^B = \sum_i (I - \text{Ad } R_i) \quad (20)$$

has a refinement-boundary defect

$$\Delta_n(X) = L_{n+1}^B j_n(X) - j_n L_n^B(X). \quad (21)$$

Its normalized boundary reduction is nonzero in all eight sectors, has 41 nonzero matrix entries and squared Frobenius norm 108/5, fills all six registered boundary-response channels, and returns to zero after the inverse crossing. Thus the stable carrier and the local generation event are distinct outputs of the same source lineage.

4.2 A finite Ward law before physical stress-energy

The boundary response, the existing IBC edge data, and the ordered-edge memory admit a six-component cell variable. If d_n is the incidence matrix of the stage path, $D_n = d_n \otimes I_6$, and H is the exact rank-six cell

Gram matrix, define

$$\mathcal{S}_n(q) = \frac{1}{2}(D_n q)^\top (I_{n-1} \otimes H)(D_n q), \quad L_n = D_n^\top (I_{n-1} \otimes H) D_n. \quad (22)$$

For every $n \geq 2$,

$$\text{rank } L_n = 6(n-1), \quad \ker L_n = \mathbf{1}_n \otimes \mathbb{F}^6, \quad (\mathbf{1}_n^\top \otimes I_6) L_n = 0. \quad (23)$$

The last identity is an exact all-finite-stage constant-shift Ward law. For every graph interval, interior edge fluxes cancel and only the two cut-boundary fluxes remain. The incidence mechanism is not uniquely Braid-specific; the Braid-specific content lies in the rank-six cell packet and its actual-versus-flip response. Equation (23) is weaker than stress-energy conservation and does not constitute a spacetime Bianchi identity.

5 Continuous cells and the native metric-affine sector

5.1 Stable-square gluing, metric faces, and curvature

The finite edge transports can be glued on stable squares with exact agreement on overlaps. On the source-native cubical cells, the generated Gram data admit continuous face representatives, and their compatible edge transports define a continuous connection with nonzero rank-four curvature on the certified face packet. A common projective timelike line and a $1 \oplus 3$ splitting can then be selected consistently on the regular sector. The induced spatial connection retains nonzero curvature.

These statements produce continuous representatives on certified finite cells; they do not derive a global smooth manifold, an arbitrary-field connection, or a physical continuum limit. A separate refinement analysis also shows why this distinction matters: the current tensor-tower embedding preserves packets but does not subdivide geometric edges, while the exact affine edge refinement is endpoint-blind to the nonuniform off-shell defects needed to select an action by refinement alone.

5.2 Torsion and nonmetricity are not removable errors

The full source connection splits into a metric-compatible polar core and a metric-symmetric part. Both curvature components scale at the same leading area order on the presently covered regular faces. Neither the tested perturbative contortion correction nor the exact finite single-face torsion-free ansatz closes. The unmodified construction therefore does not converge to the Levi-Civita connection.

This negative result selects a wider native geometry rather than eliminating gravity. Write the regular-face source coordinate as

$$\hat{K} = I - A, \quad A = UP. \quad (24)$$

With $\epsilon = \text{sign det } U$, $R = \epsilon U$, $\rho = (\det P)^{1/3}$, and $S = \rho^{-1}P$, one has the unique reconstruction

$$A = \epsilon R \rho S, \text{quad } \det S = 1. \quad (25)$$

The connected sectors have dimensions

$$\underbrace{3}_{\text{rotation}} + \underbrace{1}_{\text{dilation}} + \underbrace{5}_{\text{unimodular shear}}, \quad (26)$$

together with the discrete parity ϵ . All three continuous sectors are nontrivial in all 16 regular-face records, and their product reconstructs A exactly. This is precisely the representation type expected for a post-Riemannian affine source. The names “spin,” “dilation current,” and “shear hypermomentum” are not physical identifications here; no matter current has yet been measured or derived.

5.3 Stratified junctions

Rank-changing paths do not destroy the entire tangential metric. At the four exceptional junction records, after removal of the declared endpoint double zero, the determinant is exactly a nonzero constant times $(t - r)^2$ over $\mathbb{Q}(\sqrt{17})$, with $0 < r < 1$ in both real embeddings. At r the full Gram matrix has rank two and inertia $(0, 2, 1)$: two positive directions and one zero direction, with no negative direction. The metric is positive definite on both adjacent open intervals. The isolated event is therefore a codimension-one channel loss with a positive two-dimensional quotient, not a collapse of the tangential sector. Interpreting it as physical dimension creation would require an additional bridge and is not claimed.

6 A source-native finite first-order action

6.1 Why a metric-only action is insufficient

The full Braid boundary load decomposes into zero clock response and nonzero rotation, dilation, and shear responses. A functional depending on the spatial coframe E only through the Gram matrix $E^\top g E$ has variation in the six-dimensional image

$$\{gES : S^\top = S\}. \quad (27)$$

The rotation response lies instead in the independent three-dimensional image

$$\{gER : R^\top = -R\}. \quad (28)$$

The two images intersect only at zero, while the certified full load has nonzero components in both. Therefore the source response cannot be generated by a metric-only action. Coframe and connection variables must remain independent at the finite level.

6.2 The full-trace action

For each oriented face f , let $\mathcal{R}_f$ denote the history-dressed linear curvature obtained from the two ordered boundary paths, let E_f denote the corresponding source-generated base coframe, let $E_\top$ be the common-top coframe, and let $g_\top$ be the generated top metric. The source-native full-trace action is

$$\mathcal{S}_B[E, C, g] = \sum_{f \subset \partial c} \sigma_f \operatorname{Tr}(E_\top^\top g_\top \mathcal{R}_f[C] E_f), \quad (29)$$

where σ_f is the orientation fixed by the cell boundary. Equal column weights are forced, up to one overall scale, by covariance under the generating permutations of the three spatial columns. No face weight, sector weight, prefix-dependent scale, X51 residual, or Einstein tensor is used.

On the certified packets, the action is nonzero for both prefixes. Its analytic coframe variation agrees with exact central differences in 192/192 components, and its analytic connection variation agrees in 384/384 components. The top coframe Euler response reproduces the independently frozen full boundary solder for both prefixes, including the rotation, dilation, and shear sectors. The connection Euler response is nonzero on all 12 active edges for each prefix. Thus one finite action produces both coframe and connection equations.

The action is not a fitted scalarization of the affine translation block. Affine-origin invariance eliminates the translation scalar and selects the linear coframe-curvature pairing up to normalization. Earlier direct attempts to identify a face scalar, a Cartan three-cell scalar, or a full curvature solder with the prior X51 residual all failed by exact non-collinearity. Those no-go results are retained: Eq. (29) is justified by its own source covariance and variation, not by reverse-engineering X51.

6.3 Exact local-frame Noether identity

Let $\mathcal{E}_E$, $\mathcal{E}_C$, and $\mathcal{E}_g$ denote the coframe, connection, and top-metric Euler responses of Eq. (29). Under a local infinitesimal frame change at a vertex, the incident coframes and connections transform together. Invariance of the action gives the matrix identity

$$\mathcal{N}_v(\mathcal{E}_E, \mathcal{E}_C, \mathcal{E}_g) = 0 \quad (30)$$

at every vertex, where $\mathcal{N}_v$ is the signed sum of the coframe contribution, incident incoming and outgoing connection contributions, and, at the top vertex, the metric contribution. The identity holds exactly at all 14 non-top vertex tests and both top-vertex tests. With $g_\top$ fixed, the full $GL(4)$ identity fails as it should, while all 12 tested generators preserving $g_\top$ retain the Lorentz-type subidentity. Finite local-frame covariance also passes 16/16 tests.

Equation (30) removes the previous absence of an action-level conservation structure. It does not make the Euler responses vanish. A Noether identity relates field equations; it is not itself a solved field equation or a physical energy current.

6.4 A covariant-defect variational connector

The existing IBC, ECDR, BODC, and history-curvature library was tested as a possible independent balance action. A fixed opposite source cancels the action value, top-coframe response, and metric response at the evaluated point, but leaves eight base-coframe masks and 24 connection-edge responses. Recomputing the opposite term dynamically cancels everything only because it makes the total action identically zero. That is a tautological negative copy, not an independent physical source.

The resulting no-go did not establish that the Braid source lacks a variational connector. On an oriented edge $e = (m, d)$ with high vertex $n = m \oplus d$, define the independently generated covariant coframe defect

$$K_e = E_n - C_e E_m R_d. \quad (31)$$

This is the same X36 defect reconstructed from the source on all 24 certified edges. Its coefficient-one quadratic action is

$$\mathcal{S}_D[E, C, g] = \frac{1}{2} \sum_e \text{Tr}(K_e^\top g_n K_e). \quad (32)$$

It was frozen without reading the curvature-action Euler residual, X51, or any Einstein tensor.

The connector is nontrivial and off shell. Its exact analytic derivatives agree with central differences in 832/832 tested entries across coframe, connection, and metric slots. The induced map has rank 12 on every full defect space and rank 8 on every traversed-column-zero source subspace; the actual cotangent image is nonzero on 24/24 edges. Finite local-frame covariance holds in 16/16 tests, the infinitesimal vertex Noether identities hold in 256/256 tests, and both off-shell perturbations change the action. Thus Eq. (32) supplies the previously missing source-to-cotangent typing without fitting a negative copy.

The current result is therefore

Braid source $\rightarrow$ geometry $\rightarrow (\mathcal{S}_B, \mathcal{S}_D) \rightarrow T_{(E, C, g)}^*$,

 $\delta(\mathcal{S}_B + \mathcal{S}_D) = 0 \text{ untested.} \quad (33)$

The finite-action frontier has therefore moved. The connector no longer remains open; what remains open is whether the independently frozen coefficient-one sum $\mathcal{S}_B + \mathcal{S}_D$ makes all coframe, connection, and metric Euler responses vanish on the actual packet. Passing that test would give a finite source-balanced field equation without coefficient fitting. Failure would classify the residual sector needed for any further coupling. Neither outcome has yet been computed. The separate continuum theorem, Eq. (18), therefore remains conditional on CGR and MMR rather than being the limit of a solved finite stationary system.

7 Why scalar invariants are not enough

7.1 Soldered source holonomy

For each prefix $p \in \{A, B\}$, let

$$E_p \in \text{Mat}_{4 \times 3}(\mathbb{R}), \quad \text{rank } E_p = 3,$$

be the source-generated coframe, let $g_p \in \text{Mat}_{4 \times 4}(\mathbb{R})$ be the ambient symmetric form, and let $H_p \in GL_3$ be the internal face holonomy. Define the Gram matrix

$$M_p = E_p^T g_p E_p, \tag{34}$$

assumed invertible on the certified regular stratum.

The source-only update has two independently meaningful descriptions. The internal path mismatch is H_p , while the ambient coframe curvature is the difference between the two source-generated paths to the same face. Exact certificates show, in every one of the eight face packets, that both paths reach the same target coframe while the internal holonomy is nonidentity, and that the internal action, ambient curvature action, and source curl agree. A zero-source control fails in all eight packets.

This is not yet Riemann curvature or Einstein gravity. It is a finite source-only holonomy/curvature identity on a regular stratum. Its role here is to provide directional data that is both nontrivial and traceable to the braid source.

7.2 The scalar route and its exact obstruction

Conjugacy-invariant characteristic data of H_A and H_B provide three scalar channels per prefix. The product of their characteristic polynomials gives six nonleading coefficients that are invariant under exchange $A \leftrightarrow B$ and distinguish both one-sided collapses in the certified packets. This is a valid relational readout, but it discards direction.

The loss is not merely aesthetic. For the declared signed-permutation frame group acting on Sym_3 , the fixed subspace is one-dimensional, spanned by the identity. Therefore any full-frame-equivariant linear map from six trivially transforming scalar channels into one anisotropic Sym_3 block has rank at most one. The exact intertwiner calculation has constraint rank 30, a six-dimensional map space, and maximum anchor rank one. Moreover, the original modular residues have no unique real lift and carry no stage index.

Proposition 7.1 (Scalar-anchor no-go). *Within the declared linear, signed-permutation-equivariant anchor class, six presentation-invariant scalar channels cannot determine a full-rank anisotropic symmetric three-tensor. A covariant directional carrier must be formed before the directional information is quotiented away.*

The proposition is scoped. It does not forbid nonlinear maps, separately typed scalar observables, or enriched covariant carriers. It instead identifies the exact repair used next.

8 The generated tensor carrier on the source-derived 1 + 3 type

8.1 Construction

The formulas below were originally written in a declared $\mathbb{R}^4$ basis. In v0.5 their coordinate-free type is the source-derived carrier W of Eq. (5): $E_p \in \text{Hom}(\mathbb{R}^3, W)$, $g_p \in \text{Sym}^2(W^*)$, and $T_C \in \text{Sym}^2(W)$. A chosen

constant-plus-character basis identifies these spaces with the displayed 4×3 and 4×4 matrices. Thus the component calculations are unchanged while the numeral four is no longer a primitive target input.

Retain the holonomy defect until it has been soldered to the ambient coframe:

$$D_p = E_p(I - H_p). \quad (35)$$

Define

$$T_p = D_p M_p^{-1} E_p^\top + E_p M_p^{-1} D_p^\top \in \text{Sym}_4(\mathbb{R}), \quad (36)$$

and define the joint or intersection carrier

$$T_C = T_A + T_B. \quad (37)$$

The construction performs the quotient in the correct order. The inverse Gram matrix removes internal-basis dependence, while the coframe preserves ambient direction. Symmetrization yields a rank-two ambient carrier. Only then are the two standpoint contributions joined.

8.2 Transformation laws

Let $G_p \in GL_3$ be independent internal changes of basis. Under

$$E_p \mapsto E_p G_p, \quad H_p \mapsto G_p^{-1} H_p G_p, \quad (38)$$

we have $D_p \mapsto D_p G_p$ and $M_p \mapsto G_p^\top M_p G_p$. Substitution in Eq. (36) gives

$$T_p \mapsto T_p. \quad (39)$$

Thus the carrier does not depend on either prefix's internal presentation.

Let $L \in GL_4$ be an ambient basis change, with

$$E_p \mapsto L E_p, \quad g_p \mapsto L^{-\top} g_p L^{-1}. \quad (40)$$

Then M_p is unchanged and

$$T_C \mapsto L T_C L^\top. \quad (41)$$

This is covariance, not componentwise invariance: the object is stable while its components change tensorially.

Finally, exchanging A and B leaves Eq. (37) unchanged. Crucially, exchange invariance is not identification of the two inputs. The certified prefix carriers are distinct, and replacing the pair by (A, A) , (B, B) , or identity holonomy changes the result.

Proposition 8.1 (Structural carrier properties). *Whenever M_A and M_B are invertible, T_C is symmetric, invariant under independent internal GL_3 presentation changes, invariant under prefix exchange, and covariant by ambient congruence.*

Proof. Symmetry follows directly from Eq. (36). The internal transformation rules imply

$$D_p G_p (G_p^{-1} M_p^{-1} G_p^{-\top}) G_p^\top E_p^\top = D_p M_p^{-1} E_p^\top,$$

and similarly for the transpose term. Exchange invariance follows from commutativity of addition. Ambient covariance follows by substituting $L E_p$ and $L D_p$ into Eq. (36). $\square$

8.3 Exact finite certificate

Theorem 8.2 (Certified joint-carrier theorem). *For the registered actual source packets and their source-generated geometry:*

1. T_A and T_B are nonzero symmetric rank-three carriers;
2. T_C is nonzero, symmetric, and rank three;
3. all ten native components of $T_C \in \text{Sym}_4$ are nonzero;
4. T_C is exactly invariant under independently selected internal GL_3 gauges and under A/B exchange;
5. the congruence law in Eq. (41) holds exactly;
6. T_C differs from $2T_A$, $2T_B$, and the identity-holonomy zero control;
7. all irrational $\sqrt{17}$ parts cancel, so T_A, T_B, T_C descend exactly to rational matrices.

The theorem is an exact fixed-finite result, not a fitted numerical approximation. Items 4–6 jointly matter for the interpretation. Presentation redundancy disappears, ambient direction remains, and neither source standpoint is dispensable.

The ten components have the same mathematical type as an existing ten-component metric-response interface in the larger unification program. Equality to that response, physical normalization, units, insertion into the 87-component state, and laboratory implementation have not been proved. Accordingly, T_C is called a *carrier*, not an observed metric or stress tensor.

9 Regeneration at every finite stage

9.1 Observation mates

Let $\phi : M_5(\mathbb{C}) \rightarrow \mathbb{C}$ be any state satisfying $\phi(I_5) = 1$, and define

$$E_\phi = \text{id} \otimes \phi. \quad (42)$$

Then

$$E_\phi \circ j_n = \text{id}_{\mathcal{A}_n}. \quad (43)$$

No tracial assumption is required on the refined range. The normalized trace, a point state, and a nontracial full-support state therefore recover the same embedded source record.

For an observer word w represented by a unitary U_w , define the co-moving refinement and readback

$$J_{w,n} = \text{Ad}(U_w) \circ j_n, \quad E_{w,\phi} = E_\phi \circ \text{Ad}(U_w^{-1}). \quad (44)$$

Then $E_{w,\phi} J_{w,n} = \text{id}$.

Let $\mathcal{C}_n$ denote the deterministic composite that maps the source packet at stage n through the source-orbit constructor, geometry, holonomy, and Eq. (37) to its ten-component carrier.

Theorem 9.1 (All-finite-stage carrier naturality). *For every finite stage n , every declared observer word w , and every unital spectator-factor state ϕ ,*

$$\mathcal{C}_{n+1} \circ J_{w,n} = \mathcal{C}_n \quad (45)$$

after the co-moving readback of Eq. (44). The same carrier law and the same coefficients are used at every stage; no stage-dependent refit is permitted or required.

Proof. Equation (43) recovers each refined source component exactly. Conjugating the refinement and composing with the inverse conjugation preserves that identity. The source-orbit, geometry, holonomy, and

carrier maps are deterministic functionals of the recovered source packet, hence their composition recovers the same carrier. The exact regression checks cover eight actual source sectors, four source components, and three distinct observation states: 96/96 depth-one readbacks and 96/96 boundary co-moving re-entries. Representative depth-one and depth-two checks pass 6/6 with the same ten-channel carrier hash. $\square$

The theorem is stronger than preservation under a chosen normalized trace: the carrier is independent of which unital spectator state is used on the refined range. It is also weaker than a physical continuum theorem. No limit topology, infinite-mode Hilbert space, or continuum field is inferred from Eq. (45).

10 Main result: a counterexample to primacy

Theorem 10.1 (Finite nonterminal generated-objectivity theorem). *Under the declared source and regularity assumptions, the pointed-braid construction satisfies Definition 2.1. Specifically:*

1. *the relevant coframes and holonomies are outputs of one registered braid-source chain rather than stored target geometry;*
2. *independent internal presentations of A and B are quotiented exactly;*
3. *the joint carrier transforms tensorially under ambient changes of description;*
4. *the joint carrier is unchanged by exchanging A and B but changes when either contribution is collapsed;*
5. *the same source-native carrier law is regenerated at every finite refinement stage without fitting.*

Proof. Item 1 is the exact end-to-end source-orbit geometry result summarized in Eq. (3). Items 2–4 follow from Proposition 8.1 and Theorem 8.2. Item 5 is Theorem 9.1. $\square$

Corollary 10.2 (Failure of Objectivist Primacy as a necessity claim). *If a mathematically admissible countermodel is sufficient to test the universal claim OP, then OP is false.*

Proof. OP says that every stable, shareable, transformation-respecting carrier must be primitive. Theorem 10.1 constructs such a carrier as the nontrivial output of two maintained source standpoints and their intersection law. Therefore the universal quantifier in OP has a counterexample. $\square$

The phrase “counterexample to objectivist primacy” must be read with the antecedent in Corollary 10.2. A philosopher may reject the interpretation of the source prefixes as subjectivities, or deny that a finite formal countermodel constrains metaphysics. Those are legitimate objections. They do not invalidate the mathematics; they contest the bridge from mathematics to ontology. Conversely, the theorem does not license the stronger statement that all mind-independent reality has been disproved.

11 What is created, and what is not

11.1 Objectivity survives, but changes status

The result does not replace objectivity with private experience. The output is public in the relevant mathematical sense: internal presentation choices do not alter it, observer-word refinements admit exact co-moving recovery, and ambient changes act covariantly. What changes is the ontological order. The public structure is the result of a quotient-and-soldering operation, not the assumed stage on which that operation occurs.

This distinguishes generated objectivity from a naive averaging of perspectives. The carrier is not the common scalar part left after every direction is removed. Proposition 7.1 shows that this route is too poor to

recover an anisotropic tensor. The construction must retain the directional defect of each standpoint, solder it to a common ambient type, and only then quotient the internal descriptions. Objectivity is achieved by a disciplined relation among standpoints, not by erasing them.

11.2 The role of A , B , and C

Formally, A and B are inequivalent source prefixes with distinct certified holonomies and carriers. Their exchange is a symmetry of the joint output, but they are not identified. C is the joint carrier after internal presentation dependence has been removed. In the proposed ontology, this formal pattern is read as follows:

- A is Absolute Subjectivity, associated with light;
- B is Relative Subjectivity, an impersonal mode associated with time or a clock, not an individual, organism, mind, or conscious observer;
- C is Intersection Subjectivity, associated with space and objective manifestation.

The formal theorem establishes the grammar “distinct A and B generate a nontrivial invariant/covariant C .” It does not establish those physical nouns. Within the ontology, however, B is explicitly pre-personal and pre-biological: relative clocking and ordering are held to belong to the universe before humanity. This is the precise boundary between mathematical formulation and ontological interpretation.

11.3 Creation as regeneration rather than first manufacture

The all-stage law changes the meaning of creation. It is not a single event in which a permanent object is made once and then carried unchanged forever. At each finite stage, a refined source packet can be read back co-movingly and the same law regenerates the carrier. Thus the proposal is compatible with a world whose apparent persistence is repeated structural re-creation.

This does not mean that every stage produces a newly measured nonzero excursion. The fixed-window boundary response is nonzero in all eight certified sectors, but a nonzero boundary excursion of T_C itself has not yet been proved. The theorem establishes exact regeneration of the carrier law, not a complete physical dynamics of creation events.

12 Relation to nearby programs

Program	Shared concern	Difference from the present claim
Invariantism and structural realism	Objective content is associated with invariants or relational structure.	The present model asks whether the invariant must be primitive. Its invariant/covariant carrier is constructed only after distinct source standpoints contribute directional data.
Relational quantum mechanics	Physical facts or states are relative to physical systems rather than globally absolute.	The present result is not an interpretation of the quantum state. It constructs a specific real Sym_4 carrier and an all-stage regeneration law from a pointed-braid source.
Quantum reference frames	Descriptions transform between physical frames without an external absolute frame.	QRF work supplies precise perspective transformations within quantum kinematics. Here the primary claim concerns the generation of a shared tensor carrier from source intersection, not merely switching among descriptions of a pre-specified system.

Program	Shared concern	Difference from the present claim
QBism	The quantum state is not an objective property; interaction and agent-relative probabilities are central.	“Subjectivity” here is not belief, experience, or conscious agency. The output is an exact inter-presentation tensor, not a personal probability assignment.
Perspectival realism	Reliable knowledge can arise across a plurality of situated perspectives.	Perspectival realism is principally an account of scientific representation and inquiry. The present paper proposes a formal generative ontology and a source-level physical research program.
Braided particle models	Braid topology can encode particle-like structure and interactions [2].	The present use of braiding concerns coherence, source transport, holonomy, and generated objectivity; it does not identify Standard-Model particles with braid words.
Metric-affine gauge gravity	Coframe, independent connection, torsion, nonmetricity, hypermomentum, and Noether identities form a first-order gravitational framework [5].	The present work does not assume a continuum affine gauge theory as its microscopic source. It derives a finite $3 + 1 + 5$ affine-source type and two source-native finite actions. Under CGR and MMR it also obtains a conditional continuum Einstein equation, while both realization principles remain underived from the finite source alone.
Earlier Subjectivity Intersection work	Pointed braids provide all-orders intersection consistency and compositional objectivity [14].	The new result adds a target-free source-derived $1 + 3$ carrier, a scalar-anchor no-go, collapse separation, a no-refit all-finite-stage regeneration theorem, and the CGR/MMR conditional Einstein limit.

The novelty claim is therefore conjunctive. Neither relationality, perspective dependence, invariance, braid groups, nor process-like ontology is individually new. The contribution is the exact chain

$$\begin{aligned}
& \text{distinct braid standpoints} \longrightarrow \text{target-free } 1 + 3 \text{ carrier} \longrightarrow \text{source-only holonomy} \\
& \longrightarrow \text{direction-preserving soldering} \longrightarrow \text{joint covariant carrier} \\
& \longrightarrow \text{finite metric-affine dynamics} \longrightarrow \text{curvature and defect actions} \\
& \longrightarrow \text{local-frame Noether identities} \xrightarrow{\text{CGR+MMR}} \text{conditional Einstein equation,}
\end{aligned} \tag{46}$$

together with an explicit philosophical argument showing what this chain does and does not refute.

13 Operational content and falsification

13.1 A reproducible mathematical readout

The current operationalization is a deterministic computational readout, not a laboratory apparatus. Given the registered source packet, it performs the following steps:

1. generate the opposite source slice using the frozen signed-braid action;
2. reconstruct the affine geometry and the two source-prefix coframes;
3. compute the source-only face holonomies;
4. form D_p , M_p , T_p , and T_C by Eqs. (34)–(37);
5. output the ten independent entries of $T_C \in \text{Sym}_4$;
6. verify internal-gauge invariance, prefix-exchange invariance, ambient congruence, and the three collapse controls;

7. refine by j_n , apply a selected unital spectator state, and verify Eq. (45) without fitting.
8. construct the source graph, verify finite propagation and the boundary-event response, and evaluate the all-stage Ward kernel in Eq. (23);
9. build the continuous certified face representatives, decompose the regular source by Eq. (25), and verify the exceptional positive quotient channels;
10. fix the UB476 response before geometry, apply the three declared local-Einstein bridges, and verify the exact Riemann, Bianchi, Ricci, Einstein, and metric-two-jet identities in Eq. (14);
11. evaluate the finite action in Eq. (29), its coframe, connection, and metric variations, and the local-frame identity in Eq. (30);
12. verify the source-library balance no-go without replacing it by a fitted negative Euler residual;
13. evaluate the independent defect action in Eq. (32), its 832 exact variation checks, connector ranks, covariance tests, and Noether identities.

This is already a nontrivial readout because it returns directional tensor data that scalar invariants provably cannot supply in the declared linear equivariant class. It is not yet an empirical observable because no instrument model, units, noise model, or identification with measured geometry has been given.

13.2 Formal falsifiers

The present theorem package would fail if any of the following occurred under exact replay:

- the target geometry entered the source constructor before comparison;
- either internal GL_3 presentation changed T_C ;
- A/B exchange changed the joint carrier;
- a one-sided collapse reproduced the joint carrier;
- the ambient carrier failed the congruence law;
- the all-stage readback required a tracial spectator state or a stage-dependent coefficient;
- the source-graph support propagated faster than the certified graph bound;
- the $3 + 1 + 5$ source product failed to reconstruct the registered geometric source;
- the source-first metric two-jet failed any algebraic Riemann symmetry, Bianchi, contraction, Lorentz-signature, or exact Einstein check under its declared bridges;
- the analytic action variations failed their exact central-difference checks;
- the local-frame Noether identity failed with all required Euler slots included;
- the covariant-defect action accessed the curvature residual or an Einstein target, failed to change off shell, or lost its stated connector rank;
- any registered exact regression ceased to reproduce its frozen source record.

13.3 Physical discriminators still required

To test the Creation Principle as a claim about nature, at least one bridge must produce an empirical discriminator. Four routes are open:

1. identify T_C with, or derive a calibrated map to, an operational metric or stress response;
2. test the frozen coefficient-one sum of the curvature and defect actions, then show that any resulting stationary equations possess a controlled continuum and low-energy sector;
3. derive a boundary/re-entry signal that standard quantum-reference-frame or effective-field descriptions do not predict;
4. show that a known physical structure is recovered with fewer independent primitive assumptions and with a novel constraint among measurable channels.

“Measuring what was previously unmeasurable” would count as a genuine contribution if the new readout has an independently specified instrument, calibration, error model, and discriminating result. A renamed

internal variable would not.

14 Objections and boundaries

14.1 Where is quantum theory?

The source is quantum in a specific algebraic sense: it uses complex noncommutative matrix algebras, states, unitary pointed-braid transports, tensor refinement, and adjoint observer actions. Those ingredients are enough to make the classical-output result nontrivial, but they are current inputs. The paper does not derive complex amplitudes, the Born rule, Hilbert-space composition, unitary time evolution, or quantum-field locality from a bare topological braid group. Accordingly, “a route to quantum structure” names the wider Braid-first programme. The exact result of this paper begins at a quantum-algebraic Braid source and derives the declared classical carrier and finite gravity dynamics from it.

14.2 Where is spacetime, and does the ambient four-space smuggle objectivity back in?

Version 0.5 removes the numerical and ambient-carrier forms of this objection at the mathematical layer. The finite source produces $Q_{\text{obs}} = V_4$. Full affine-observer symmetry, an order unit, and point separation force the complete readout and hence the target-free split $1 \oplus 3$. Observer Uniform Cauchy Closure completes that carrier to $W = \mathbb{R}^{V_4}$ without adding a direction; $\text{Aff}(W)$ is then its own smooth four-dimensional affine carrier. The source path is not renamed as physical three-space, and the four metric-response controls are not renamed as region coordinates.

What remains conditional is physical realization, not the numeral four. CGR must identify the affine carrier with local physical events, solder the source-net regions and Cartan data to its tangent/cotangent fibers, and supply the continuum first-variation limit. The cup/null Lorentz form is dimensionless and also requires physical interpretation and calibration. Thus a completed objective four-dimensional container is no longer a primitive mathematical input, while curved physical spacetime is still not claimed as a source-only theorem.

14.3 Is the carrier unique?

No global uniqueness theorem is claimed. Equations (35)–(37) are a natural, exact, source-native solution that obeys the declared symmetries and controls. Other nonlinear or higher-order carriers may exist. The counterexample to OP needs existence, not uniqueness. A stronger physical theory would benefit from deriving the carrier from an action, universality principle, minimality theorem, or experimental constraint.

14.4 Is a finite model enough?

It is enough to refute a universal mathematical necessity claim if the model is admitted. It is not enough to establish a fundamental theory of our universe. The nonterminal tower prevents the result from being only one isolated finite face, while the source graph and continuous certified cells add finite notions of locality and curvature. They do not by themselves produce a global continuum spacetime. CGR and MMR supply a conditional local continuum neighborhood and infrared Einstein equation, not a source-only continuum theorem. Unitary quantum-field evolution, renormalized stress, global extension, and empirical scale identification remain separate tasks.

14.5 Where is gravity?

The construction contains more than prerequisites for gravity. Equations (8) and (9) give an exact source-native coframe update and covariant curvature closure, while Eq. (10) gives the corresponding finite metric-affine update. Rank-changing geometry requires the source term, and every nonmetricity, source-linear, and source-quadratic contribution survives ablation. The same packet extends without refitting through all finite stages. The source graph supplies finite locality and a Ward law; continuous certified faces carry nonzero connection curvature; Eq. (29) supplies an exact first-order curvature action with Eq. (30) as its local-frame identity; and Eq. (32) supplies an independent off-shell variational connector. In this exact fixed-orbit sense, an action-bearing finite Braid metric-affine gravity candidate has been constructed.

The relation to Einstein gravity has two levels. First, the independently fixed source generates a local Lorentz metric two-jet satisfying Eq. (14) exactly at one point under three explicit bridges. Second, the off-shell action (16) yields Eq. (18) on a neighborhood under CGR and MMR. The second statement is a genuine conditional variational derivation, not merely formal similarity or a fitted metric jet. What remains open is derivation of CGR and MMR from the Braid source, finite combined stationarity, physical units, quantization of the continuum gravitational field, and empirical identification. The noninvolutive four-flow result excludes one first-class route but does not negate the single local GKSL evolution. The paper therefore claims a Braid-based quantum-gravity candidate with a two-principle conditional Einstein limit, not completed quantum gravity or source-only general relativity.

14.6 Is “subjectivity” only a relabeling?

It would be a relabeling if the formal roles made no difference. Here the roles are constrained: two prefixes must remain distinct; their internal presentation can be quotiented but their contributions cannot be collapsed; and their intersection produces a carrier with a verified transformation law. This is more than terminology. Nevertheless, calling these formal roles “subjectivities” is an interpretive move. The mathematics fixes the relational structure; ontology assigns its meaning.

15 The Braid-first Creation Principle as a research programme

The present work suggests a sequence of progressively stronger claims:

1. **Quantum-origin adequacy:** quantum operational structure is derived from less assumed Braid data rather than placed in the source algebra. This remains open.
2. **Formal generation:** distinct Braid standpoints generate exact classical objective structure. This paper establishes that claim for the declared quantum-algebraic source.
3. **Finite dynamical adequacy:** the source law generates exact coframe and metric-affine updates, a finite local net, curvature-bearing continuous cells, a curvature action, and an independent covariant-defect action with exact local-frame Noether identities. This paper records that result.
4. **Conditional pointwise Einstein adequacy:** under three explicit bridges, the independently fixed source generates a Lorentz metric two-jet satisfying $G_{\mu\nu} = (3/5)\Theta_{\mu\nu}$ exactly at one point. This paper establishes that bounded arrival result.
5. **Conditional continuum Einstein adequacy:** under CGR and MMR, the continuum metric-affine action yields $G_{\mu\nu} = (3/5)T_{\mu\nu}$ as a neighborhood Euler–Lagrange equation. This paper establishes the conditional derivation; Braid-only derivation of the two realization principles remains open.
6. **Stationary-field adequacy:** independently frozen source functionals balance every coframe, connection, and metric Euler slot without target fitting. The required connector now exists, but exact stationarity of the coefficient-one combined action has not yet been tested.
7. **Physical-sector adequacy:** the generated carrier, action, and affine-source sectors map to calibrated metric, stress-energy, hypermomentum, or other operational fields. This remains open.

8. **Unconditional continuum and low-energy adequacy:** the source law itself generates the bridge assumptions, a global local-causal spacetime dynamics, and established low-energy gravity. The conditional local equation is obtained here, but this stronger source-only claim remains open.
9. **Empirical selection:** a measurement distinguishes the Creation Principle from lower-commitment relational or effective descriptions. This remains open.
10. **Ontological adoption:** the Creation Principle is accepted as the best account of what reality is. This would require the preceding physical and empirical work plus philosophical argument; it cannot be obtained from the finite certificates alone.

This hierarchy is not a retreat from the ontological thesis. It states what would make that thesis scientifically serious. The first level changes the dialectical situation: objectivist primacy can no longer be defended merely by asserting that no exact alternative can produce stable public structure. The remaining dispute concerns physical realization, explanatory economy, and evidence.

16 Conclusion

Subjectivity-Intersection Braid Quantum Gravity asks whether quantum structure, spacetime, and gravity can be recovered from one standpoint-bearing braid law. Its exact finite core is now substantial. A quantum-algebraic pointed-braid source with two distinct standpoints generates nontrivial directional holonomies. If these are reduced too early to scalar invariants, an exact rank obstruction prevents reconstruction of a full anisotropic tensor. If instead each holonomy defect is soldered through its source-generated coframe, internal presentation dependence disappears while ambient direction survives. Their sum yields a real rational Sym_4 carrier that is exchange-invariant, covariant, nonzero, collapse-separated, and reproducible by one law at every finite stage of a nonterminal tower.

The same source lineage produces a target-free $1 + 3$ carrier, exact finite coframe and metric-affine Braid gravity equations, a source-local causal graph and GKSL evolution, continuous certified faces with nonzero connection curvature, a canonical $3 + 1 + 5$ post-Riemannian source decomposition, and curvature and defect actions with exact local-frame identities. The pointwise result then constructs an Einstein-compatible Lorentz metric two-jet from an independently fixed source response. The variational result goes further: under CGR and MMR, the continuum action (16) yields the neighborhood field equation (18). The present dependency chain is therefore

quantum-algebraic standpoint-bearing Braid source $\longrightarrow$ generated objective carrier
 $\longrightarrow$ finite metric-affine dynamics and actions
 $\longrightarrow$ pointwise Einstein arrival
 $\xRightarrow{\text{CGR+MMR}}$ conditional continuum Einstein equation.

This enlarged chain strengthens rather than displaces the refutation of objectivist primacy. The counterexample no longer generates only a stable shared tensor. It generates finite local geometry, curvature, variational response, and, conditionally, an Einstein field law on continuous spacetime. Objectivity survives as covariance, reproducibility, and inter-presentation stability; what fails is the universal claim that such objective structure must be ontologically prior to every standpoint relation. This is a constructive refutation of objectivist *primacy*, not a denial of objective knowledge and not an empirical disproof of mind-independent reality.

The quantum-gravity claim remains bounded. CGR and MMR have not been derived from the Braid source, and the continuum gravitational field has not been quantized. The current temporal rank-four response distribution is noninvolutive, although the combined local GKSL evolution exists. Ultraviolet completion, physical scale, and a discriminating natural-data prediction remain open. The Creation Principle makes

the stronger ontological proposal that objective manifestation is continually generated when distinct subjectivities intersect without becoming identical. The present paper supplies that proposal with a target-free $1 + 3$ mathematical carrier, an exact finite foundation, and a two-principle conditional continuum gravitational extension; its physical truth remains open to assumption removal and empirical selection.

Data, code, and provenance

The exact certificates used in this paper are stored in the SIEL Research Agent repository. The v0.2 carrier and finite-gravity chain is preserved at commit `3d1cccc86`. The post-v0.2 locality, conditional local-Einstein, continuous-cell, metric-affine, action, Noether, source-balance, and covariant-defect connector lineage is preserved through commit `27dd70da0`. The first continuum derivation and its historical four-law audit are preserved in BGCE038. BGCE075–BGCE127 preserve the temporal rank, non-closure, action, normalization, KMS, inductive-limit evolution, and quasilocal-locality results. BGCE128 removes the primitive numerical and ambient four-carrier input at the mathematical layer, and BGCE129 proves the minimal current CGR/MMR dependency packet. These records are preserved through commit `75d2c2691db515e35ef8bdec82047b0c1d56306f`. Principal records include A57S-X62–X99, A57S-CE1A–CE4W, and BGCE001–BGCE129; superseded implementation failures remain preserved and are not counted as scientific outcomes. The repository is [SIEL-Research/SIEL-Research-Agent](#). The gate records contain frozen protocols, exact or high-precision raw outputs as appropriate, result classifications, verifiers, endpoint-provenance checks, and manifests. The reported results are exploratory theoretical and computational results; they are not independent replication, peer review, or empirical validation.

Declaration on computational assistance

OpenAI Codex was used under the author’s direction for mathematical ideation, exact-check scripting, audit organization, literature triage, and drafting. The author retains responsibility for definitions, interpretation, verification, citations, and any decision to release or submit the work.

References

- [1] E. Artin, “Theory of braids,” *Annals of Mathematics* **48**(1), 101–126 (1947). [doi:10.2307/1969218](#).
- [2] S. O. Bilson-Thompson, “A topological model of composite preons,” arXiv:hep-ph/0503213 (2005). [doi:10.48550/arXiv.hep-ph/0503213](#).
- [3] C. A. Fuchs and R. Schack, “Quantum-Bayesian coherence,” *Reviews of Modern Physics* **85**, 1693–1715 (2013). [doi:10.1103/RevModPhys.85.1693](#).
- [4] F. Giacomini, E. Castro-Ruiz, and Č. Brukner, “Quantum mechanics and the covariance of physical laws in quantum reference frames,” *Nature Communications* **10**, 494 (2019). [doi:10.1038/s41467-018-08155-0](#).
- [5] F. W. Hehl, J. D. McCrea, E. W. Mielke, and Y. Ne’eman, “Metric-affine gauge theory of gravity: field equations, Noether identities, world spinors, and breaking of dilation invariance,” *Physics Reports* **258**, 1–171 (1995). [doi:10.1016/0370-1573\(94\)00111-F](#).
- [6] J. Ladyman, “What is structural realism?” *Studies in History and Philosophy of Science Part A* **29**(3), 409–424 (1998). [doi:10.1016/S0039-3681\(98\)80129-5](#).
- [7] M. Massimi, *Perspectival Realism*. Oxford University Press (2022). [doi:10.1093/oso/9780197555620.001.0001](#).
- [8] M. Massimi, “Epistemic communities and their situated practices: Perspectival realism—a primer,” *Annals of the New York Academy of Sciences* **1523**, 5–10 (2023). [doi:10.1111/nyas.14981](#).

- [9] T. Nagel, *The View from Nowhere*. Oxford University Press (1986).
- [10] C. Rovelli, “Relational quantum mechanics,” *International Journal of Theoretical Physics* **35**, 1637–1678 (1996). doi:[10.1007/BF02302261](https://doi.org/10.1007/BF02302261).
- [11] A. Vanrietvelde, P. A. Höhn, F. Giacomini, and E. Castro-Ruiz, “A change of perspective: switching quantum reference frames via a perspective-neutral framework,” *Quantum* **4**, 225 (2020). doi:[10.22331/q-2020-01-27-225](https://doi.org/10.22331/q-2020-01-27-225).
- [12] R. M. Wald, *General Relativity*. University of Chicago Press (1984).
- [13] H. Weyl, *Symmetry*. Princeton University Press (1952).
- [14] S. Watanabe, *Subjectivity-Intersection Mathematics: All-Orders Intersection Consistency and Compositional Objectivity from Pointed Unitary Braids*, version 1.0, Zenodo (2026). doi:[10.5281/zenodo.22166167](https://doi.org/10.5281/zenodo.22166167).
- [15] S. Watanabe, *Subjectivity Intersection Ontology: A Comparative Ontological Framework for Third Subjectivity*, version 1.0.0, Zenodo (2026). doi:[10.5281/zenodo.21641878](https://doi.org/10.5281/zenodo.21641878).
- [16] S. Watanabe, *Subjectivity-Intersection Mathematics Public Working Preprint v4.2 Draft*, Zenodo (2026). doi:[10.5281/zenodo.21909933](https://doi.org/10.5281/zenodo.21909933).